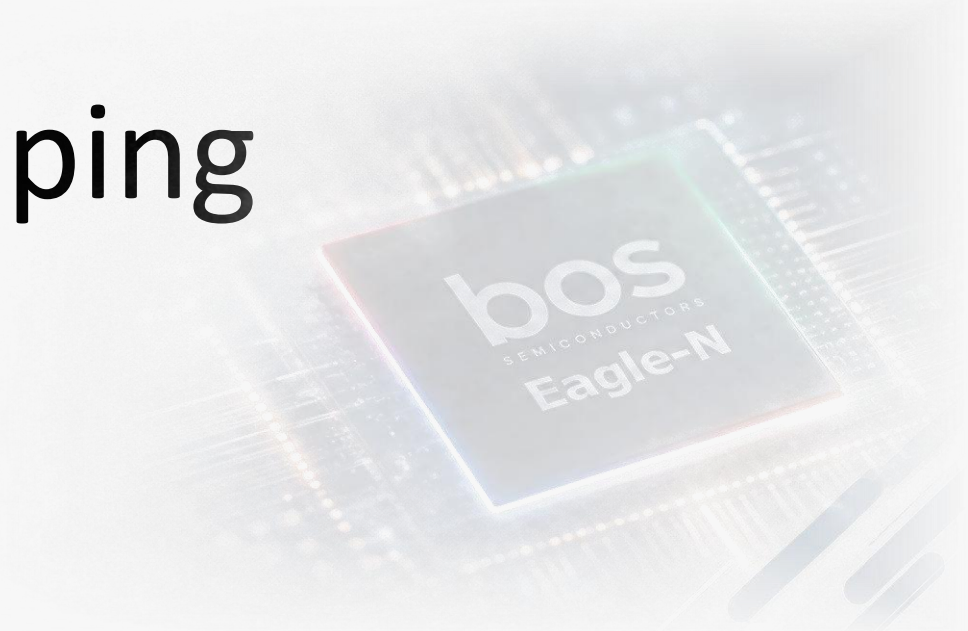
2026.8.21.

Aging Aware NPU Remapping

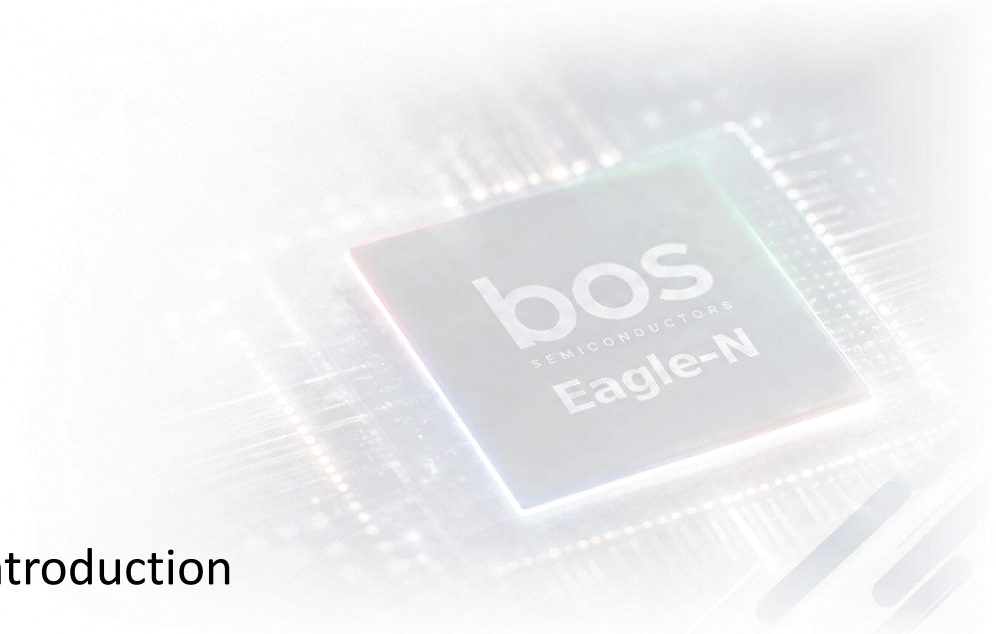


N P U : N e v e r P a u s e U n i t

박준범, 윤수민, 최민기, 최윤성

bos SEMICONDUCTORS x 성균관대학교 SUNG KYUN KWAN UNIVERSITY

Index



01 Introduction

02 Reliability simulation

03 Conv core remapping API

04 Limitations and Future work

Automtive IC Reliability

Long-term operation

Thermal & Electrical stress

Safety-critical system

2020 Toyota Prius NHTSA recall

266,637 vehicles recalled

Repeated High-load Driving

↓

Higher Thermal Stress in Specific IPM Transistors

↓

Transistor Damage Over Time

↓

Hybrid System Shutdown → Loss of Motive Power

AEC-Q100

Appendix Template 4A: AEC-Q100 Qualification Test Plan

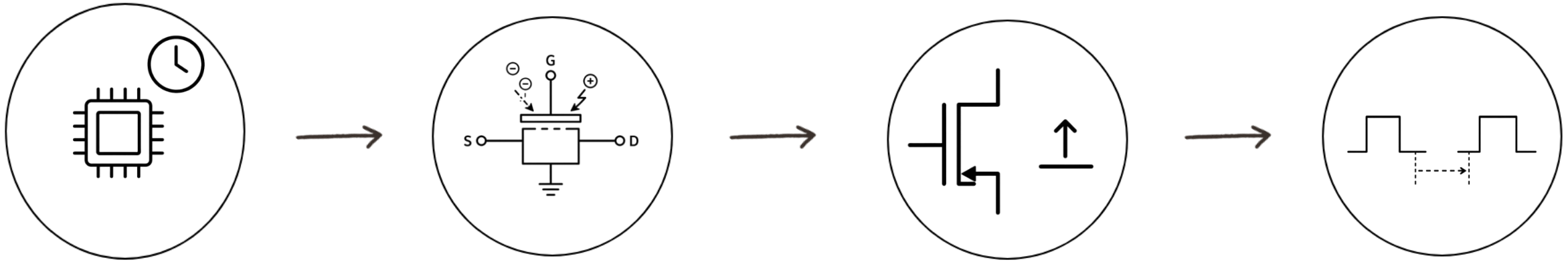
Q100J QUALIFICATION TEST PLAN

USER COMPANY:				DATE:			
USER P/N:				TRACKING NUMBER:			
USER SPEC #:				USER COMPONENT ENGINEER:			
SUPPLIER COMPANY:				SUPPLIER MANUFACTURING SITES:			
SUPPLIER P/N:				PPAP SUBMISSION DATE:			
SUPPLIER FAMILY TYPE:				REASON FOR QUALIFICATION:			
STRESS TEST	**ABV**	**TEST#**	**TEST METHOD**	**Test Conditions/S.S. per Lot/# Lots (identify temp, RH, & bias)**	**REQUIREMENTS S.S**	**# LOTS**	**RESULTS Fails/S.S./# lots**
Preconditioning	PC	A1	JEDEC J-STD-020 JESD22-A113	Peak Reflow Temp. = Preconditioning used =	Min. MSL = 3		MSL =
Temperature Humidity Bias or HAST	THB / HAST	A2	JESD22-A101/A110		77	3	
Autoclave or Unbiased HAST or Temperature Humidity	AC / UHST / TH	A3	JESD22-A102/A118/A101		77	3	
Temperature Cycle	TC	A4	JESD22-A104		77	3	
Power Temperature Cycling	PTC	A5	JESD22-A105		45	1	
High Temperature Storage Life	HTSL	A6	JESD22-A103		45	1	
High Temperature Operating Life	HTOL	B1	JESD22-A108		77	3	
Early Life Failure Rate	ELFR	B2	AEC Q100-008		800	3	
NVM Endurance, Data Retention, & Operational Life	EDR	B3	AEC Q100-005		77	3	
Wire Bond Shear	WBS	C1	AEC Q100-001		5	1	
Wire Bond Pull Strength	WBP	C2	MIL-STD-883 - 2011		5	1	
Solderability	SD	C3	J-STD-002		15	1	
Physical Dimensions	PD	C4	JESD22-B100/B108		10	3	
Solder Ball Shear	SBS	C5	AEC Q100-010		10	3	
Lead Integrity	LI	C6	JESD22-B105		5	1	
Bump Shear	BST	C7	JESD22-B117		5	1	
Electromigration	EM	D1					
Time Dependent Dielectric Breakdown	TDDB	D2					
Hot Carrier Injection	HCI	D3					
Bias Temperature Instability	BTI	D4					
Stress Migration	SM	D5					
Pre- and Post-Stress Electrical Test	TEST	E1	Test to spec				
ESD - Human Body Model	HBM	E2	AEC Q100-002				
ESD - Charged Device Model	CDM	E3	AEC Q100-011				
Latch-Up	LU	E4	AEC Q100-004		3	1	
Electrical Distributions	ED	E5	AEC Q100-009		30	3	
Fault Grading	FG	E6	AEC-Q100-007			1	
Characterization	CHAR	E7	AEC Q003				
Electromagnetic Compatibility	EMC	E9	SAE J1752/3		1	1	
Short Circuit Characterization	SC	E10	AEC Q100-012		10	3	
Soft Error Rate	SER	E11	JESD89-1, -2, -3		3	1	
Lead Free	LF	E12	Q005				
Process Average Test	PAT	F1	AEC Q001				
Statistical Bin/Yield Analysis	SBA	F2	AEC Q002				
Hermetic Package Tests	MECH	G1-4	Series		15	1	
Package Drop	DROP	G5			5	1	
Lid Torque	LT	G6	MIL-STD-883 - 2024		5	1	
Die Shear Strength	DS	G7	MIL-STD-883 - 2019		5	1	
Internal Water Vapor	IWV	G8	MIL-STD-883 - 1018		5	1	
Supplier:				Approved by: (User Engineer)			

Solderability	SD	C3	J-STD-002
Physical Dimensions	PD	C4	JESD22-B100/B108
Solder Ball Shear	SBS	C5	AEC Q100-010
Lead Integrity	LI	C6	JESD22-B105
Bump Shear	BST	C7	JESD22-B117
Electromigration	EM	D1	
Time Dependent Dielectric Breakdown	TDDB	D2	
Hot Carrier Injection	HCI	D3	
Bias Temperature Instability	BTI	D4	
Stress Migration	SM	D5	
Pre- and Post-Stress Electrical Test	TEST	E1	Test to spec
ESD - Human Body Model	HBM	E2	AEC Q100-002

AEC, *AEC-Q100 Rev. J1, Failure Mechanism Based Stress Test Qualification for Integrated Circuits in Automotive Applications*, 2026.
NHTSA, *Part 573 Safety Recall Report 20V-369, Toyota Motor Engineering & Manufacturing*, 2020.

Semiconductor Aging



Long-Term Operation

BTI-Induced Aging

Threshold Voltage Shift

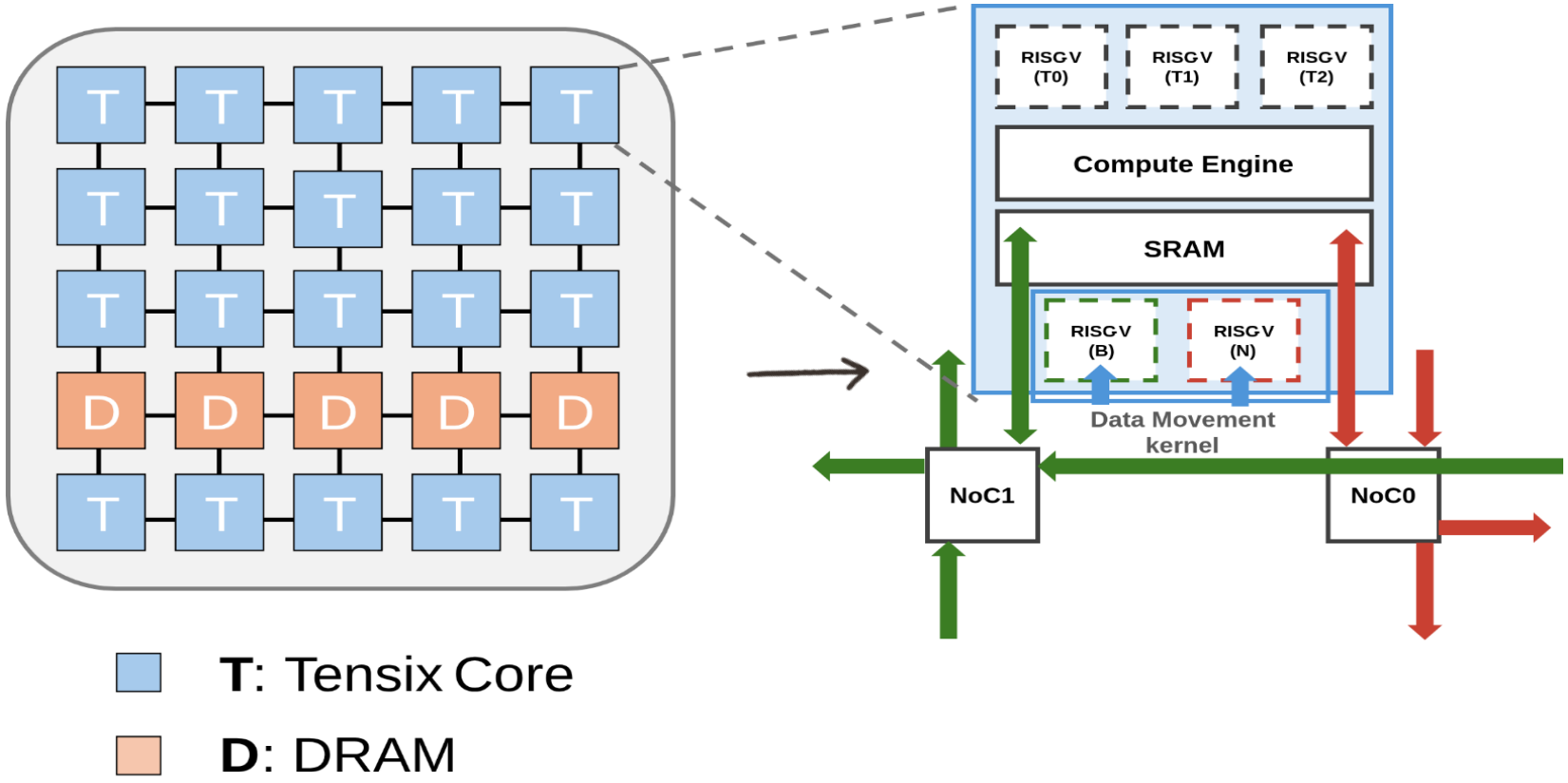
Circuit Delay Increase

NBTI

$$\Delta V_{th,NBTI} = A_{NBTI} \times e^{\left(\frac{-k_G}{kT}\right)} \times V_{dd}^{4} \times t^{1/6} \times u^{1/6}$$

X. Yang et al., “Development and Challenges of Reliability Modeling From Transistors to Circuits,” IEEE Journal of the Electron Devices Society, vol. 11, pp. 179–189, 2023.

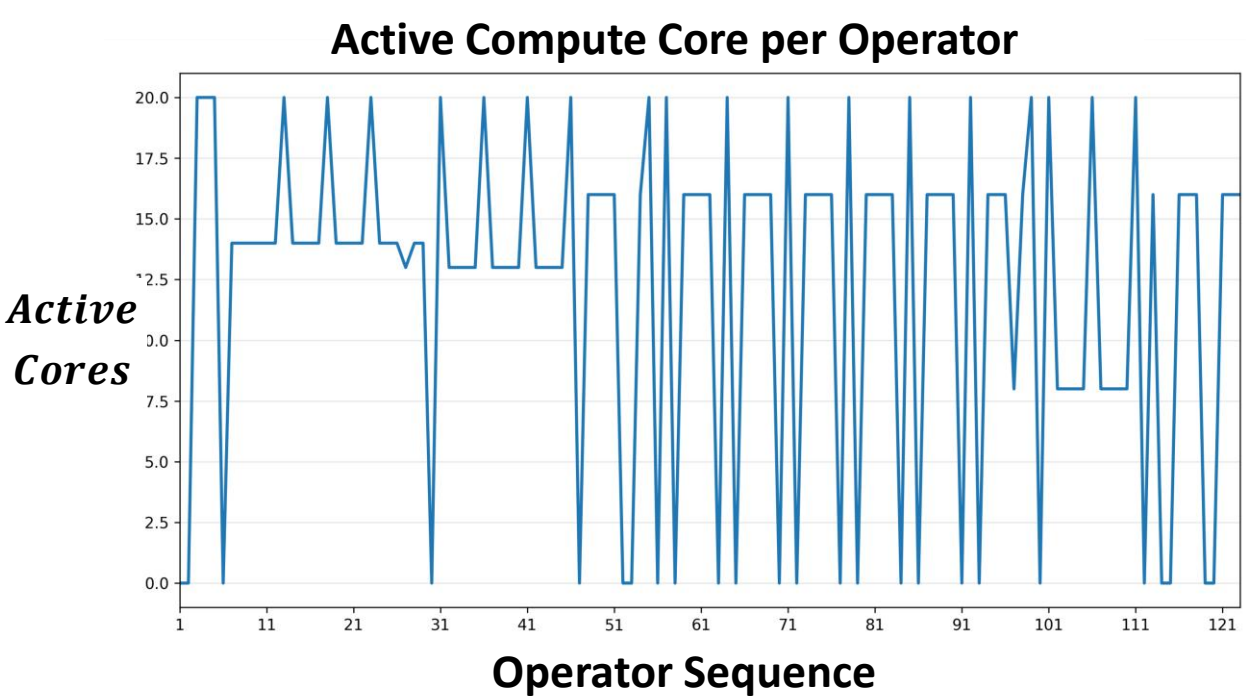
BOS Eagle-N A0 Architecture

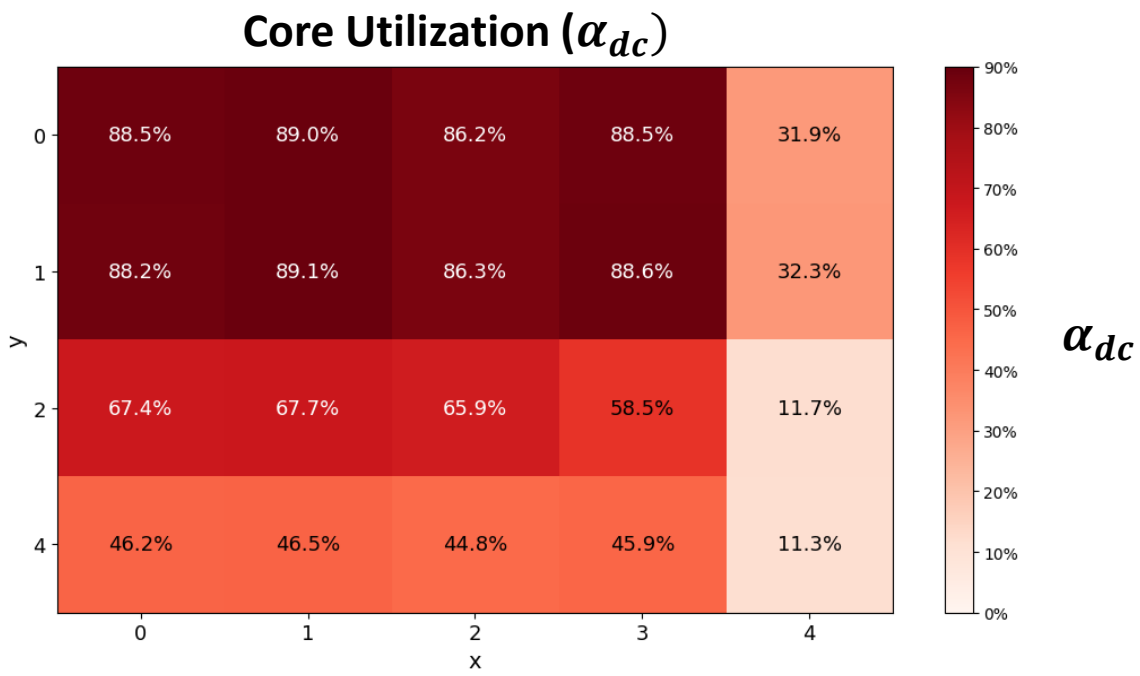


Multi-core Tensix architecture interconnected via NoC

BOS Semiconductors, “Eagle-N, AI Accelerator for ADAS & IVI.”
Tenstorrent, TT-Metalium Documentation, “Compute Engines and Data Flow within Tensix.”

Measured Core Utilization Imbalance





Core utilization differs by up to 7.92× across the compute array.

- Acc (Classification Accuracy) – Empirical Model

$$Acc(RRMSE) = \frac{(1 + e^{-ms})(Acc_{clean} + nc^{-1})}{1 + e^{s(RRMSE - m)}} + nc^{-1}$$

m and s : fitting parameters characterizing error resilience.

nc : number of classification types(resnet50: 1000)

RRMSE: The relative proportion of the error scaled to the magnitude of the original data.

H. Huang et al., "Statistical Modeling of Soft Error Influence on Neural Networks," in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, vol. 42, no. 11, pp. 4152-4163, Nov. 2023, doi: 10.1109/TCAD.2023.3266405.

- **Measurement:** Random bit error injection into weights / Measure m, s

BER	Acc(Faulty)	RRMSE
1e-07	0.7957	0.0001
1e-06	0.7956	0.0017
1e-05	0.7957	0.0245
5e-05	0.7969	0.0274
0.0001	0.7955	0.0490
0.001	0.7555	0.3210
0.002	0.7645	0.3401
0.003	0.7367	0.3753
0.005	0.5642	0.6878
0.007	0.5703	0.5861
0.01	0.5532	0.6358
0.013	0.3058	0.8526
0.016	0.2384	0.9022
0.02	0.0757	1.0195
0.03	0.0029	1.2041
0.05	0.0000	1.0890
0.1	0.0000	1.1108

Table. ResNet-50 Error Injection Test (N=10,032)

Fitting Results
Acc(Clean) :
0.7957
m : 0.7620
s : 7.5978

- Estimate BER vs. RRMSE relationship.
- Regression fitting using Weibull-like Exponential Saturation $y = C \cdot \left(1 - e^{-k \cdot x^b}\right)$

- Bit Error Rate - Statistical approximation of the timing violation probability

$$BER_{raw}(t) = P\big(\text{Margin} < \Delta delay(t)\big) = \Phi\left(\frac{\Delta delay(t) - \mu_{margin}}{\sigma_{margin}}\right)$$

$\Delta delay(t)$ = delay increase at time t, μ_{margin} = mean guardband, σ_{margin} = guardband standard deviation

A. Agarwal, D. Blaauw and V. Zolotov, "Statistical timing analysis for intra-die process variations with spatial correlations," ICCAD-2003. International Conference on Computer Aided Design (IEEE Cat. No.03CH37486), San Jose, CA, USA, 2003, pp. 900-907, doi: 10.1109/ICCAD.2003.159781.

- NBTI-Induced Delay

$$\frac{\Delta\text{delay}(t)}{d_{\text{nominal}}} = x_i(t) = \kappa \cdot \Delta V_{th,i}(t) = \kappa C \cdot S_i(t)^n$$

$S_i(t)$: Cumulative stress time
Approximate that delay variation is linearly proportional to threshold voltage variation.

- Assume a 10% delay increase over T_0 years.

$$\kappa C \cdot T_0^n = 0.10 \quad \rightarrow \quad \kappa C = \frac{0.10}{T_0^n}$$

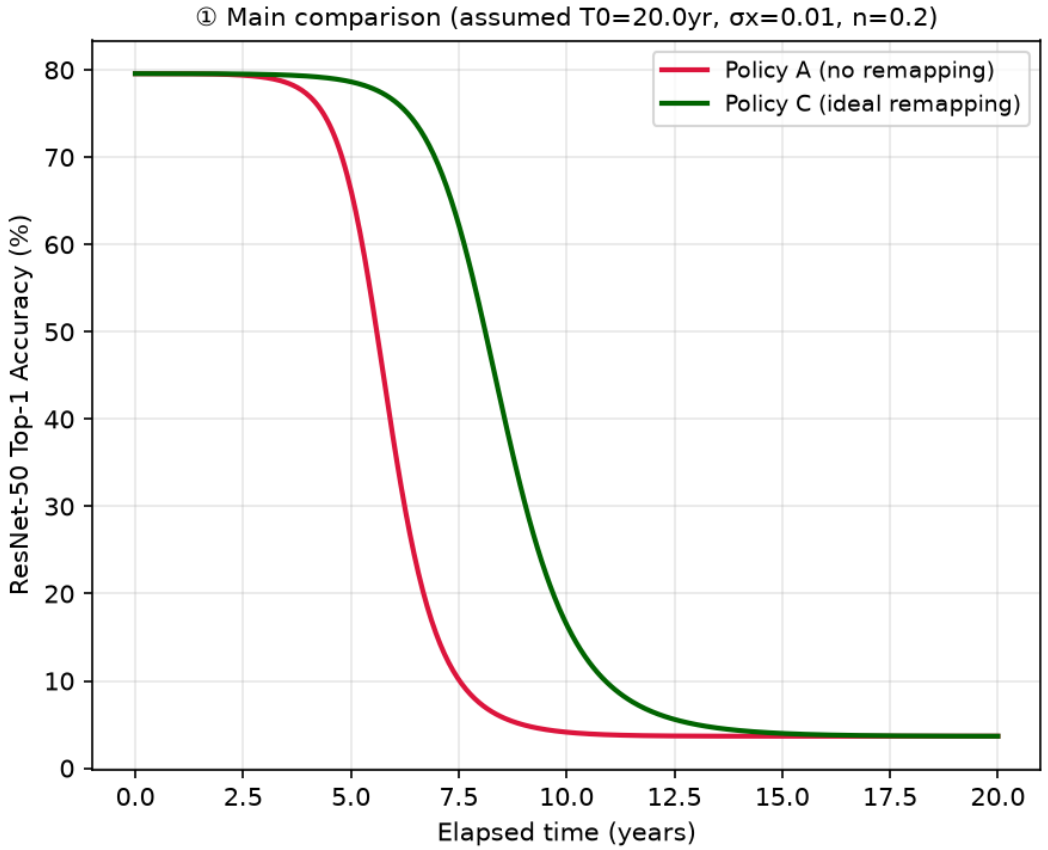
- Final Model Formula

$$x_i(t) = \kappa C \cdot S_i(t)^n = \left(\frac{0.10}{T_0^n}\right) \cdot S_i(t)^n = 0.10 \cdot \left(\frac{S_i(t)}{T_0}\right)^n$$

$$BER_i(t) = \Phi\left(\frac{x_i(t) - 0.10}{\sigma_x}\right) = \Phi\left(\frac{0.10 \cdot \left[\left(\frac{S_i(t)}{T_0}\right)^n - 1\right]}{0.01}\right)$$

V. Veetil, Y. -H. Chang, D. Sylvester and D. Blaauw, "Efficient smart monte carlo based SSTA on graphics processing units with improved resource utilization," Design Automation Conference, Anaheim, CA, USA, 2010, pp. 793-798, doi: 10.1145/1837274.1837474.

- Result



A difference of ~ 2 years

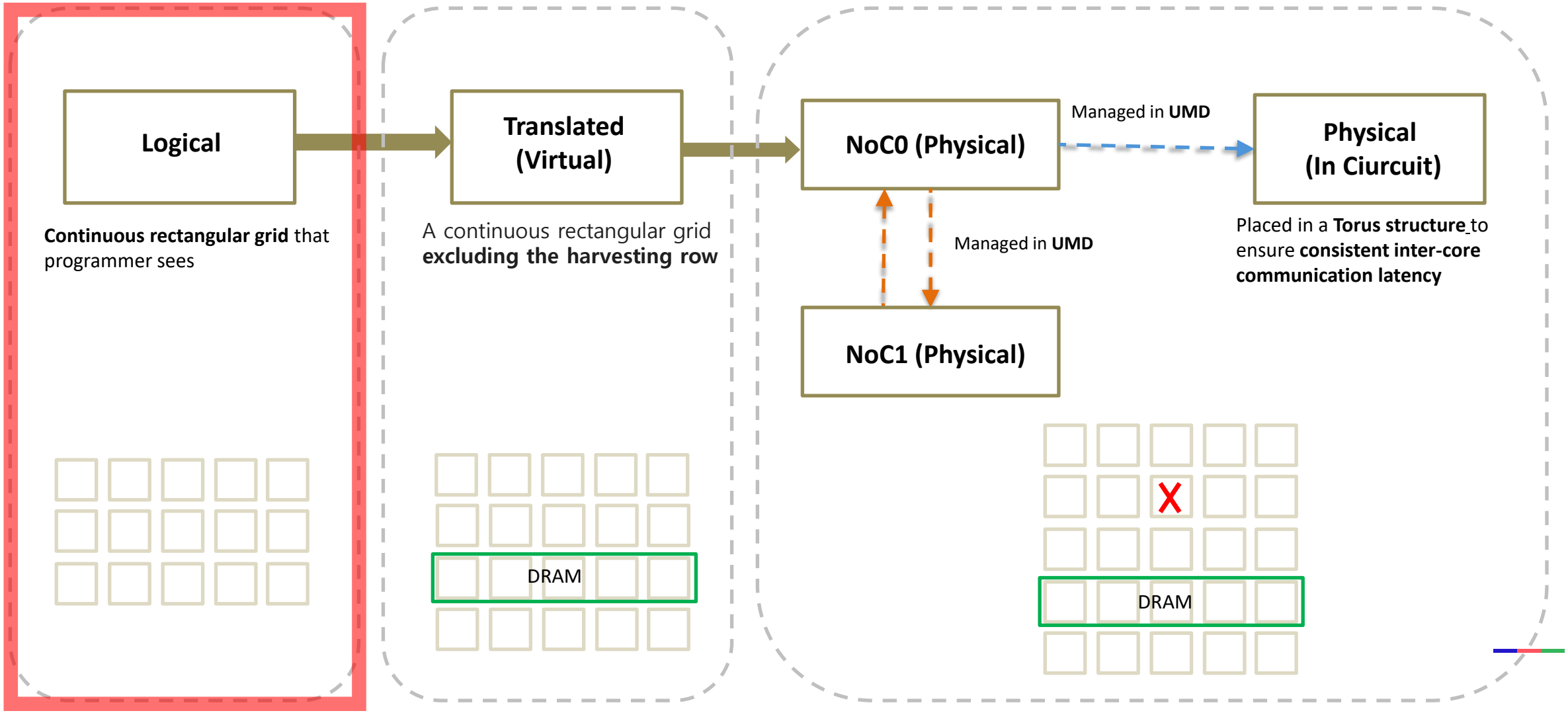
Conv Core Remapping API



01 Background & Motivation
02 Problem Statement
03 Core Remapping
04 Remapping Planner
05 Profile-Guide Optimization
06 Evaluation & Conclusion

01 Background & Motivation

Tenstorrent Coordinate System (X, Y)

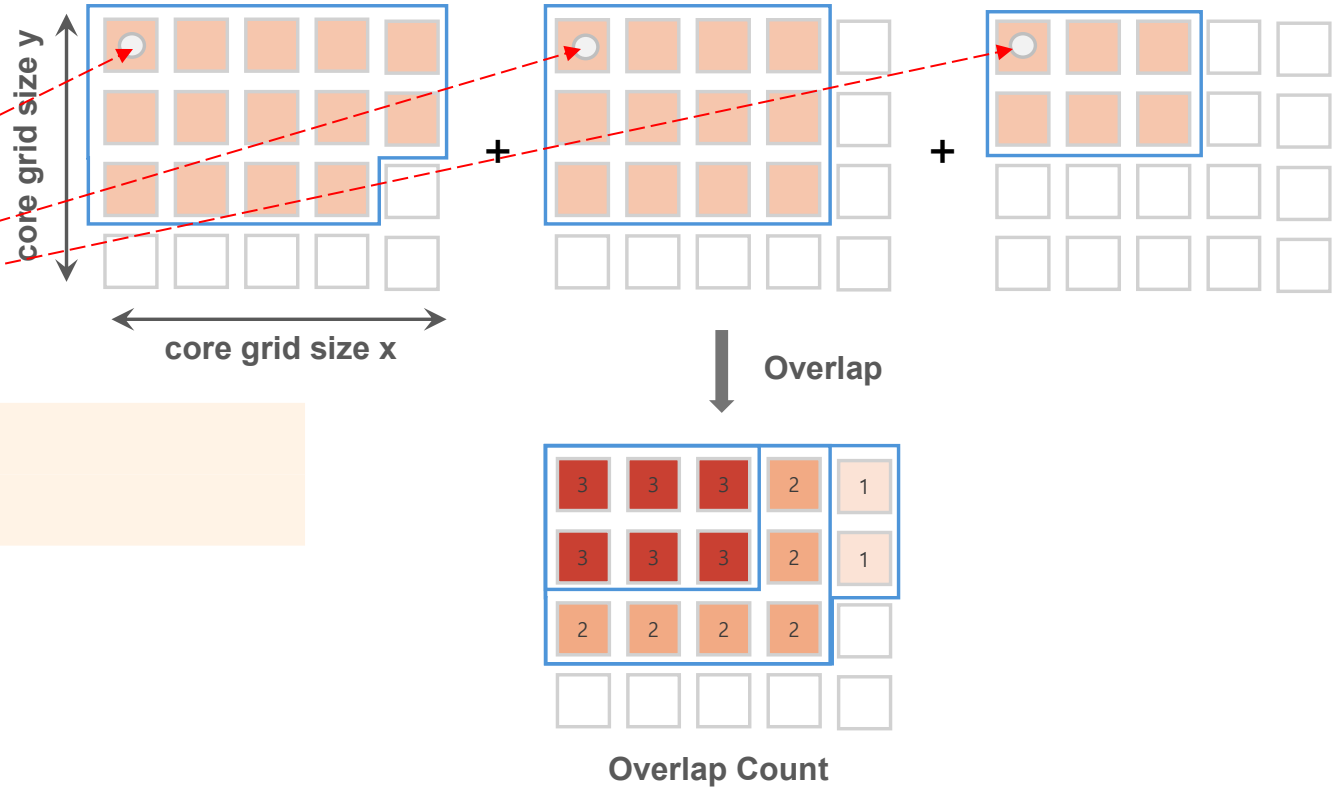


01 Background & Motivation

```
uint32_t cores_x = ...;
uint32_t cores_y = ...;

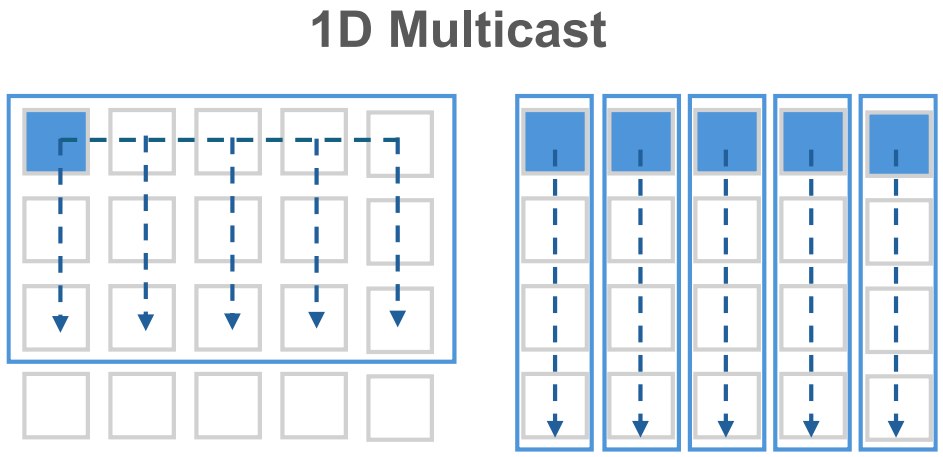
CoreRange core_range = CoreRange(
    CoreCoord({0, 0}),
    CoreCoord({cores_x - 1, cores_y - 1})
);

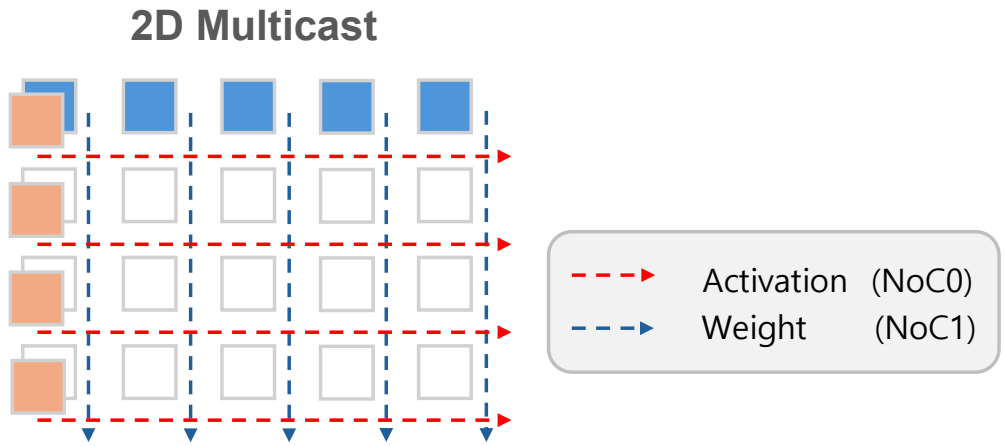
grid = CoreRangeSet({core_range});
```



Grid size is operation-dependent, but its origin is fixed.

02 Problem Statement – Spatial Data Movement on Convolution





-A single sender broadcasts data to all participating receivers.

-The sender in each column independently multicasts along one axis.

- Activations are multicast row-wise over NoC0, while weights are multicast column-wise over NoC1.

-Because they use separate NoC, the two data streams are transferred in parallel.

02 Problem Statement – Spatial Data Movement on Convolution

```
get_noc_multicast_addr(
    x_start, y_start,
    x_end,   y_end,
    l1_addr);
```

-A single sender broadcasts data to all participating receivers.

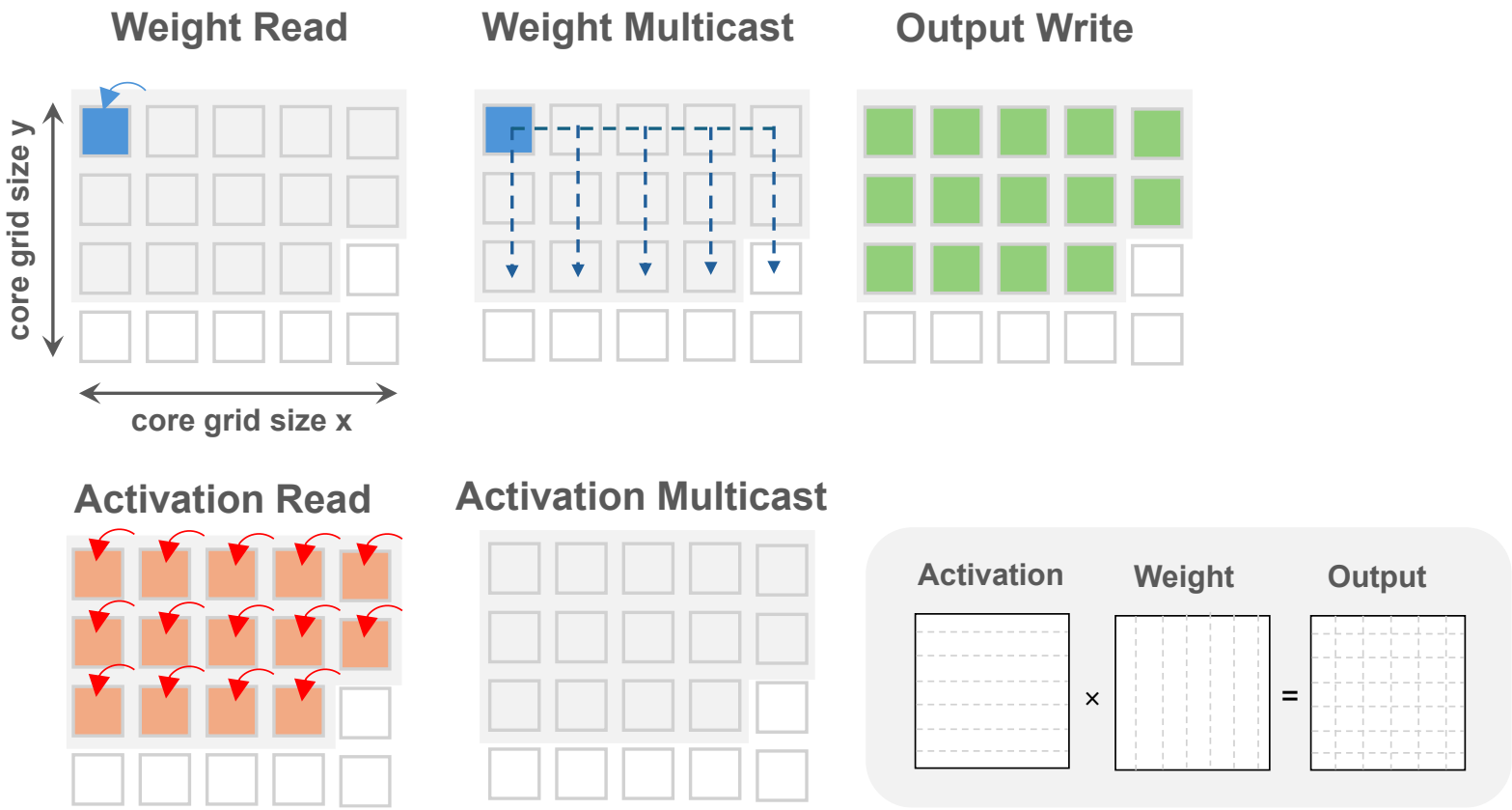
-The sender in each column independently multicasts along one axis.

- Activations are multicast row-wise over NoC0, while weights are multicast column-wise over NoC1.

-Because they use separate NoC, the two data streams are transferred in parallel.

02 Problem Statement – Spatial Data Movement on Convolution

Height Sharding Data Movement (DRAM Read & Multicast)



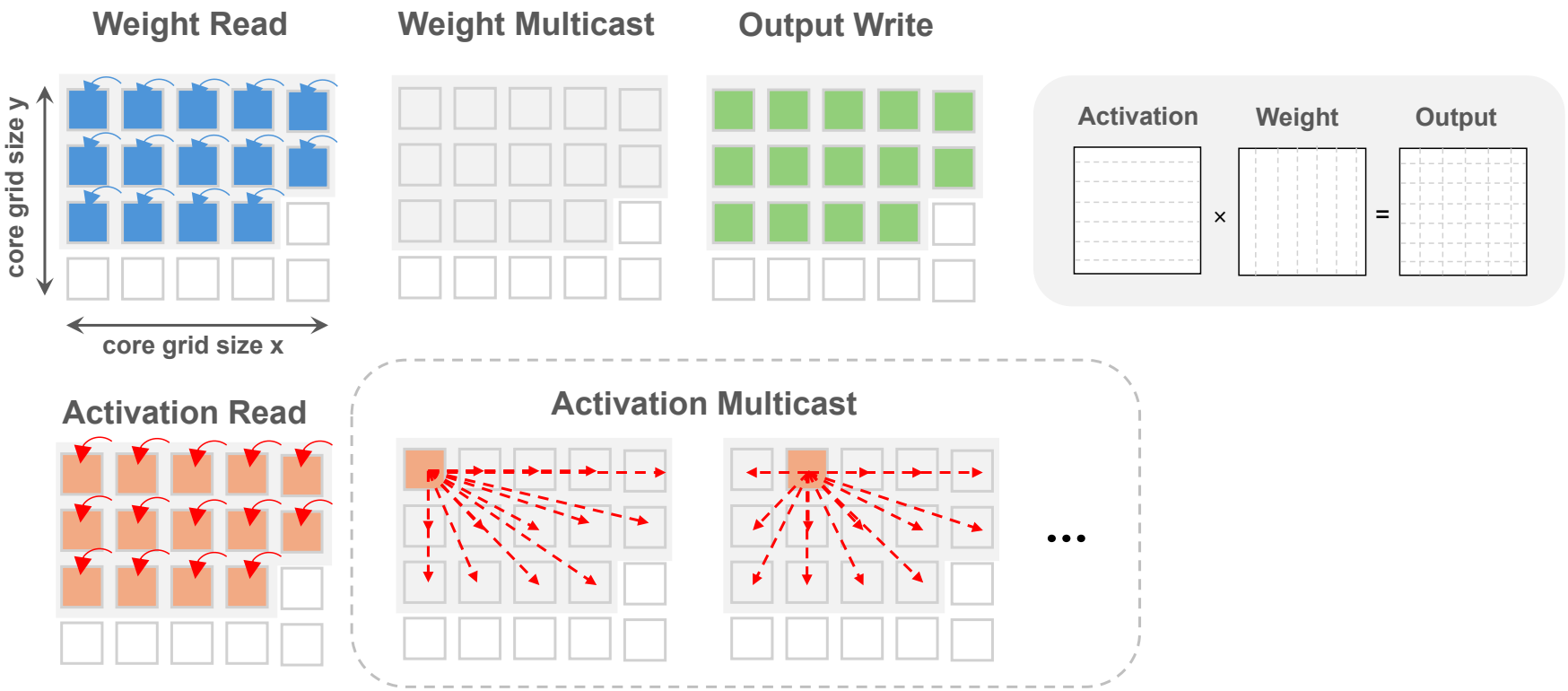
02 Problem Statement – Spatial Data Movement on Convolution

Block Sharding Data Movement (DRAM Read & Multicast)



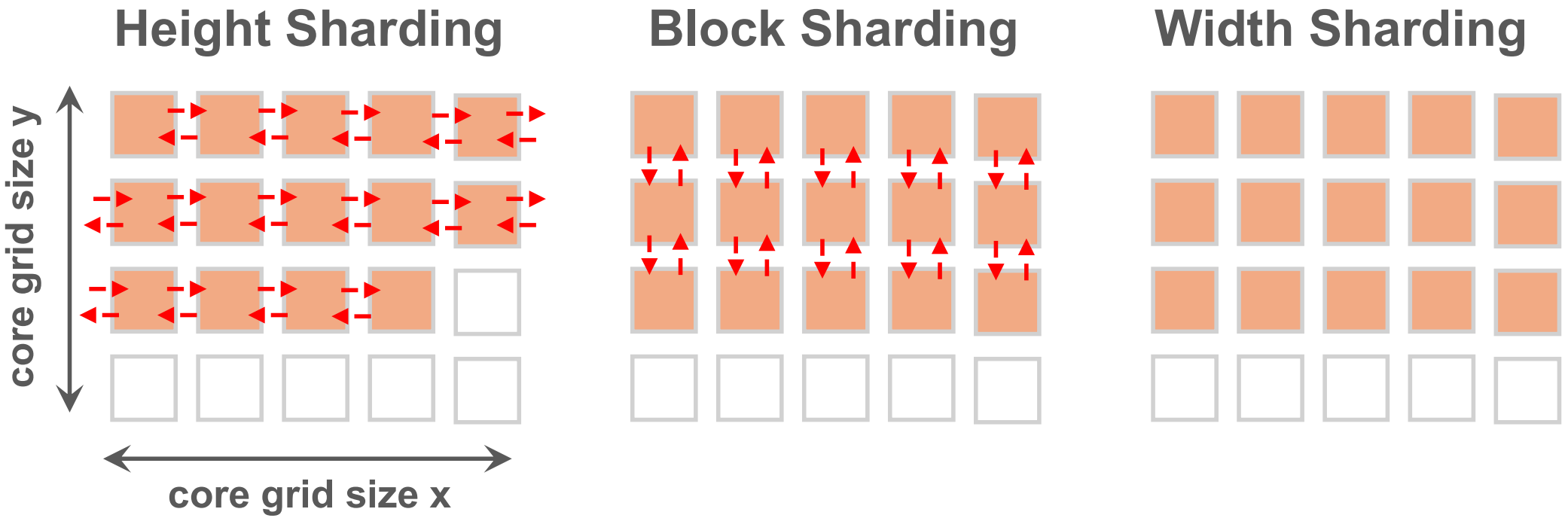
02 Problem Statement – Spatial Data Movement on Convolution

Width Sharding Data Movement (DRAM Read & Multicast)

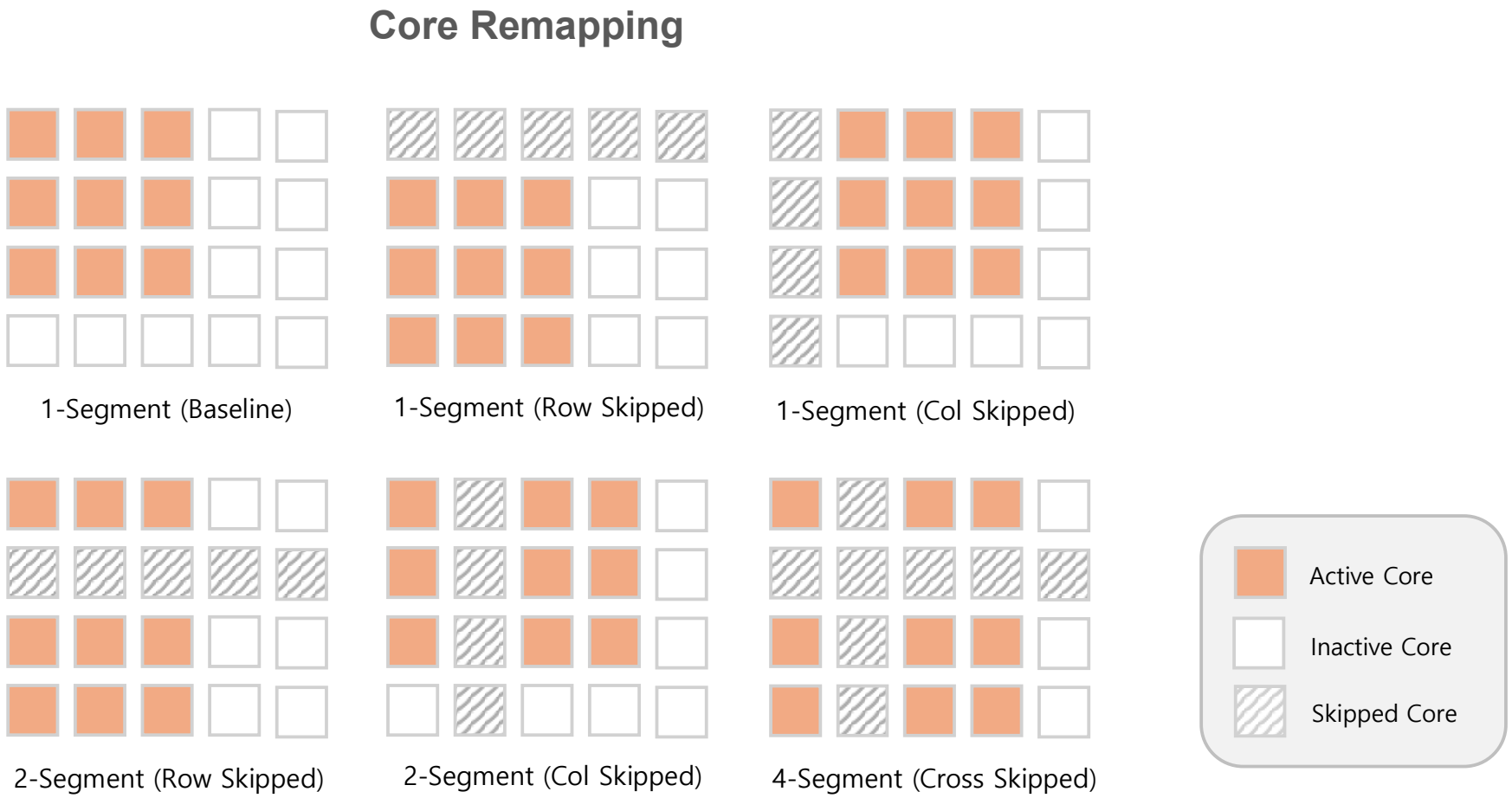


02 Problem Statement – Spatial Data Movement on Convolution

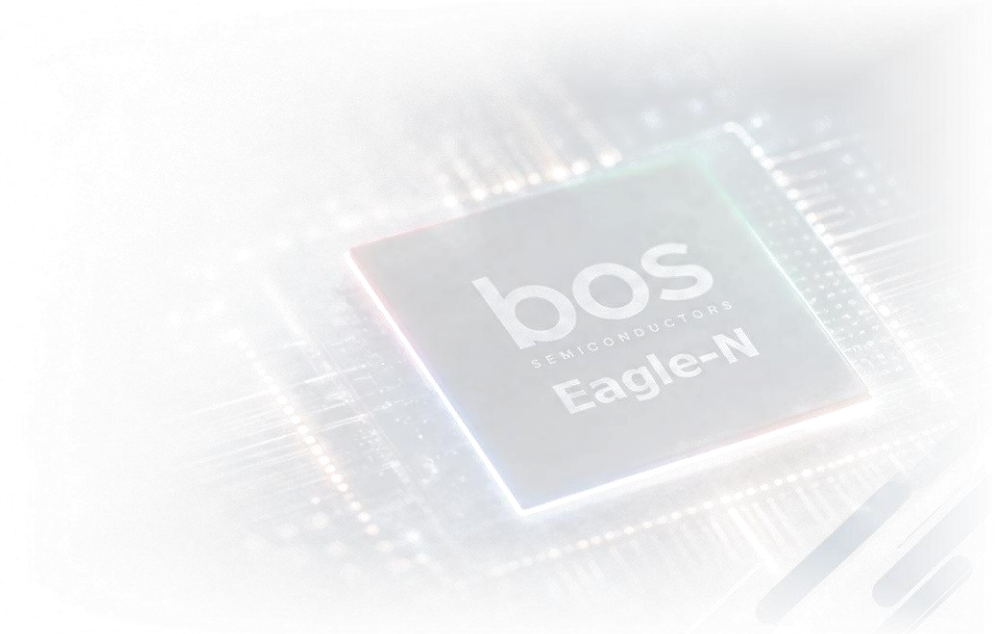
Haloing Data Movement



02 Problem Statement – Core Remapping Strategy on Convolution



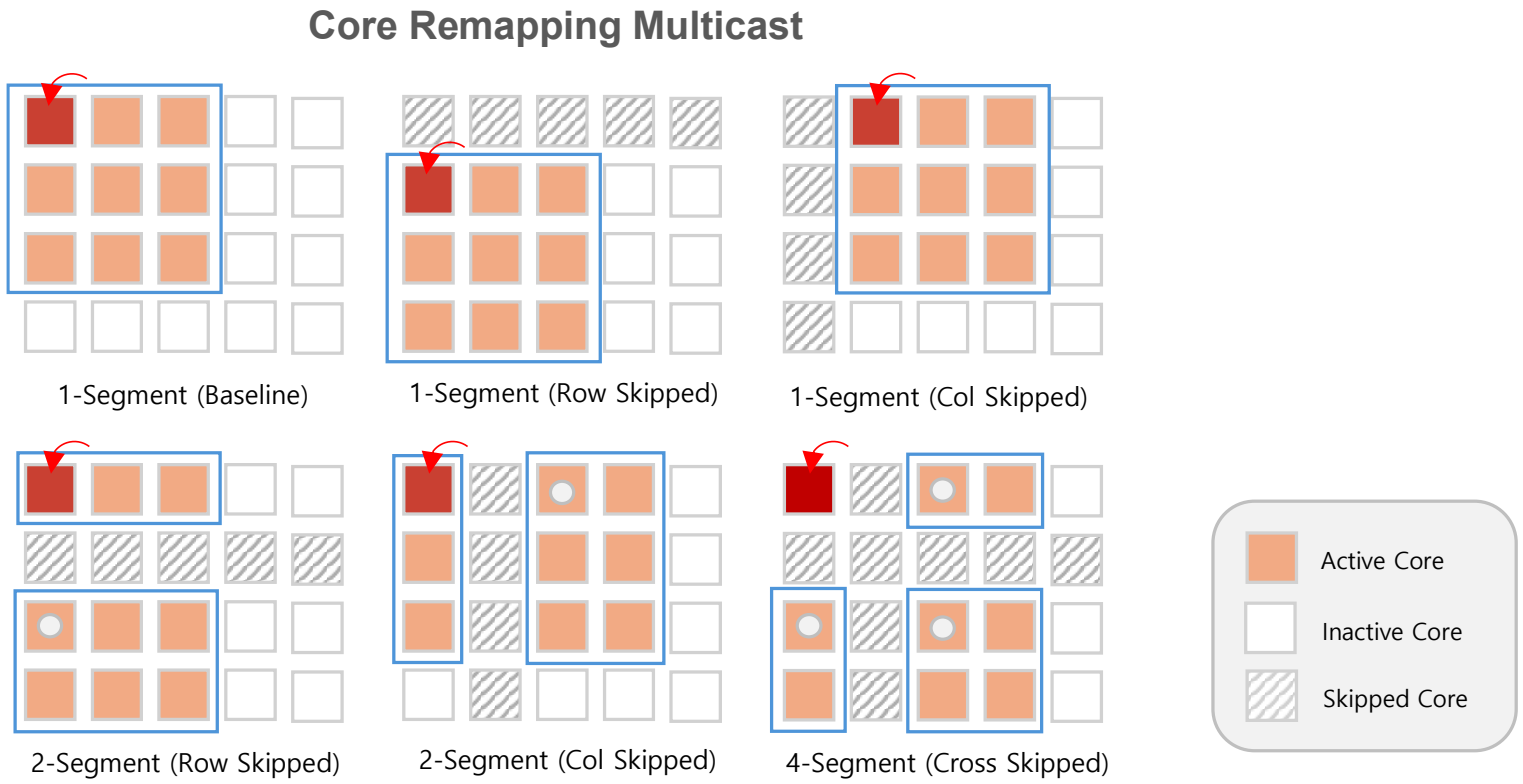
03 Core Remapping



03 Core Remapping

Row/Column Exclusion API and Multicast Regions

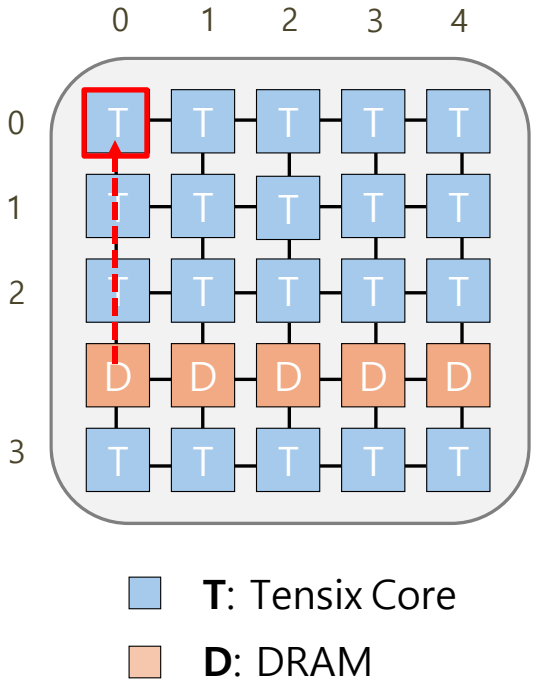
```
config = ttnn.Conv2dConfig(
    enable_core_remapping=True,
    excluded_worker_rows=[1],
    excluded_worker_columns=[2],
)
```

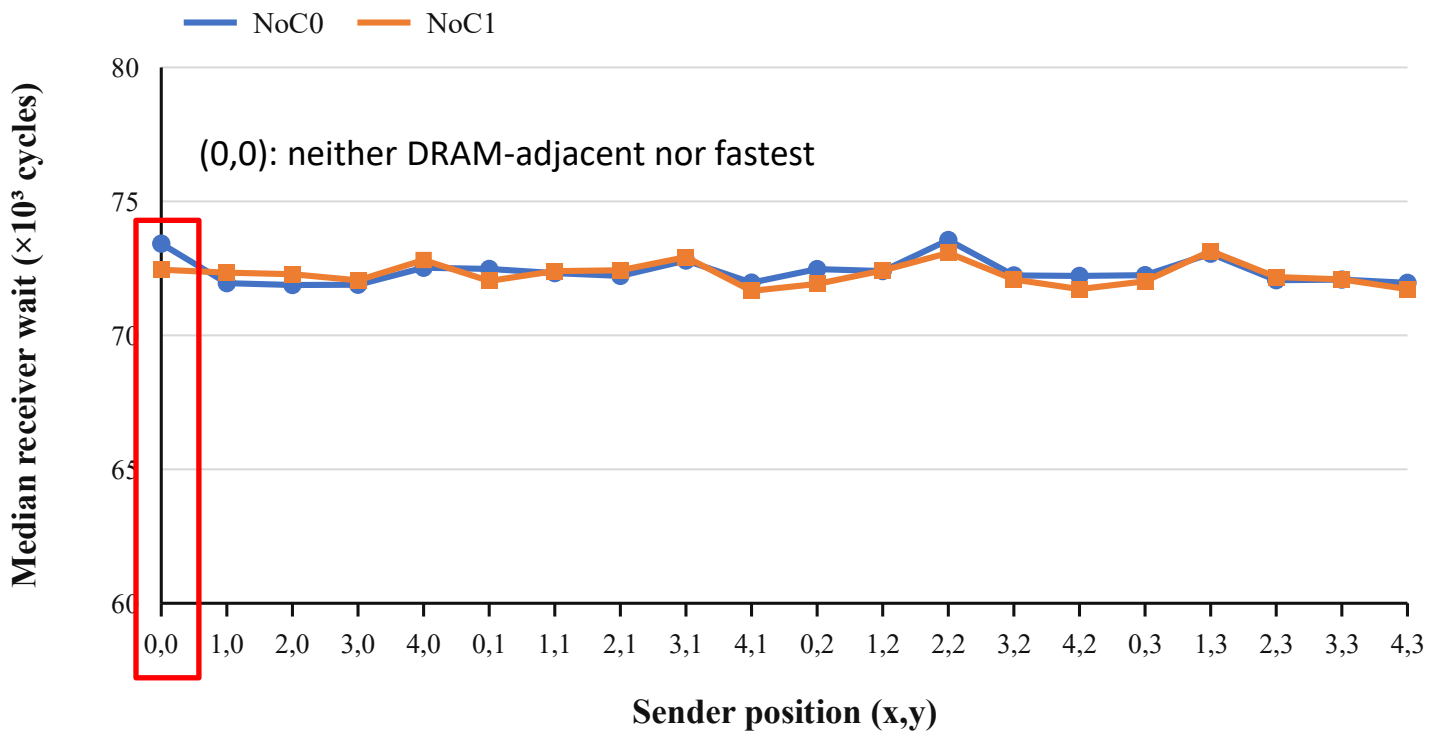


What is the cost of moving and splitting the worker grid?

03 Core Remapping – The Cost of Data Movement

Sender Placement Has Limited Impact on 5×4 Multicast Completion

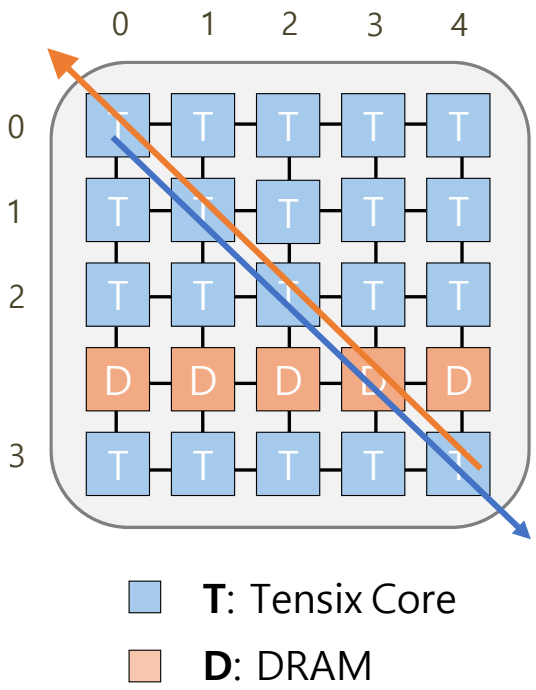


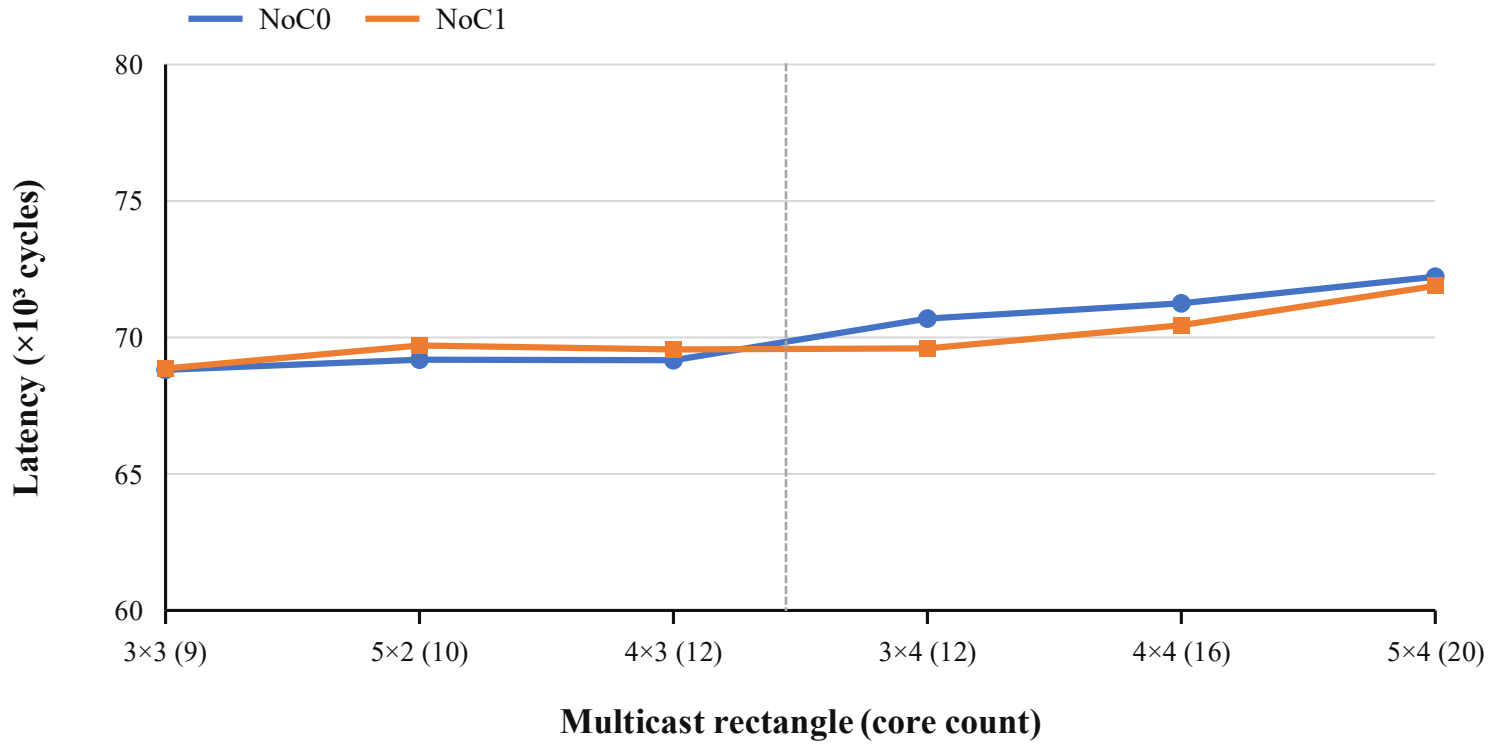


Moving the sender across all 20 cores changes median receiver wait by only about 2%.

03 Core Remapping – The Cost of Data Movement

Latency Generally Increases with **Core Count**

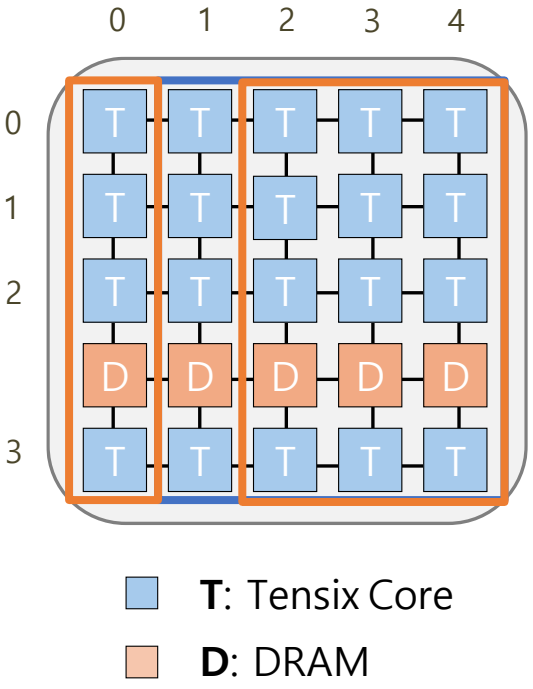


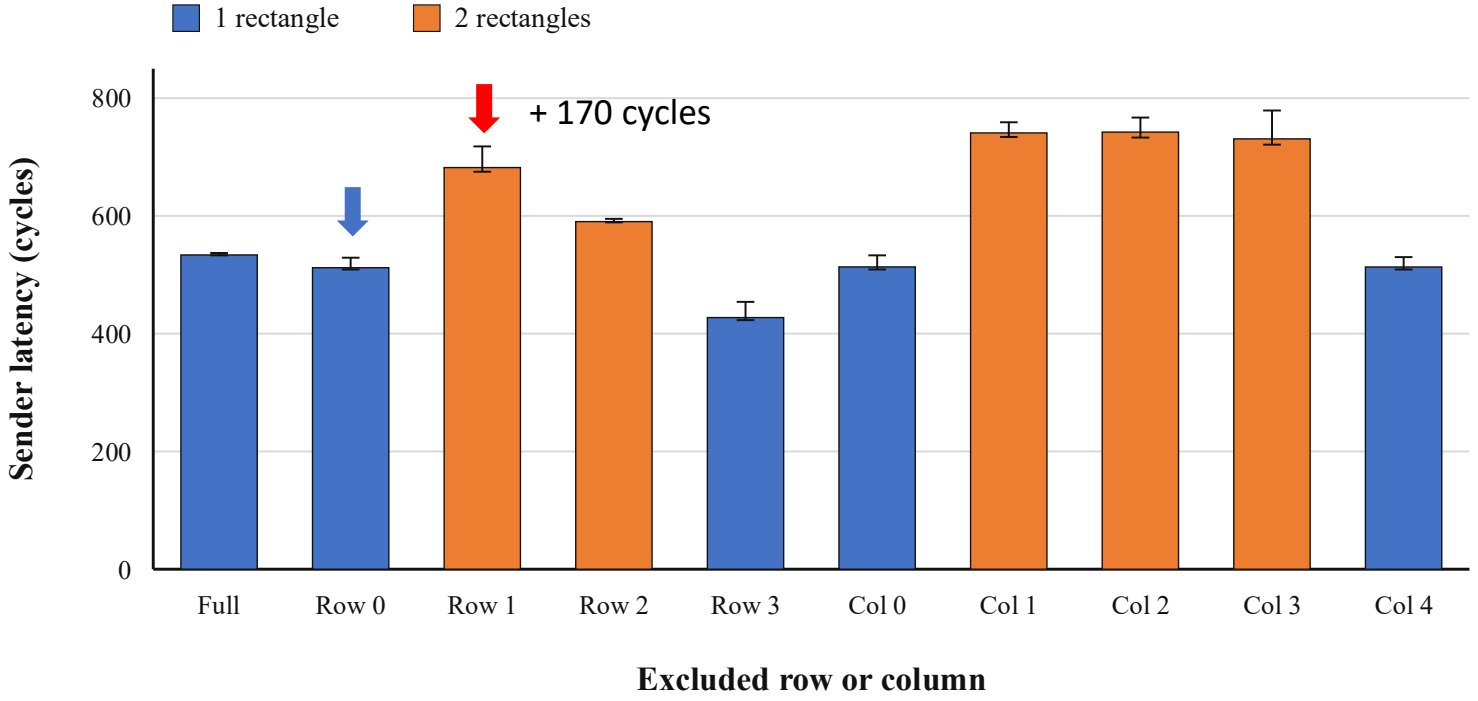


Increasing destinations from 9 to 20 cores raises completion time by **4.4–5.0%**.

03 Core Remapping – The Cost of Data Movement

Disconnected Placements Require More Multicast Work

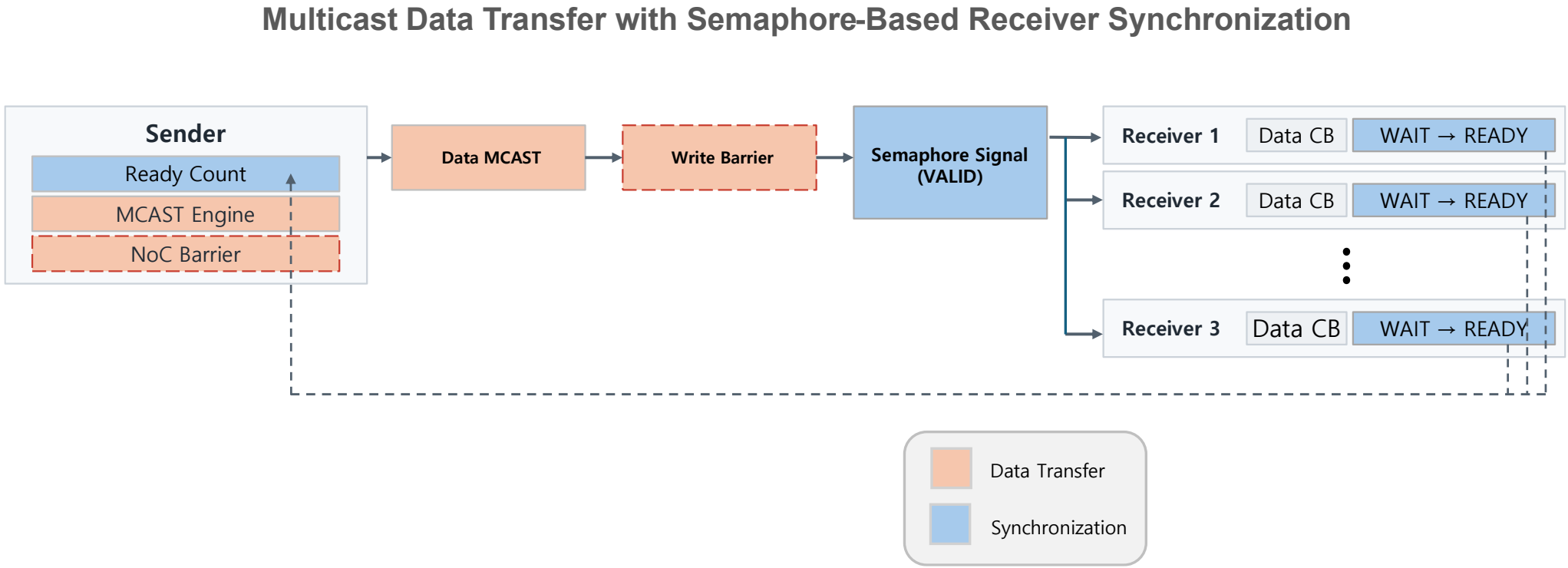




Each disconnected region requires a separate rectangular multicast command.

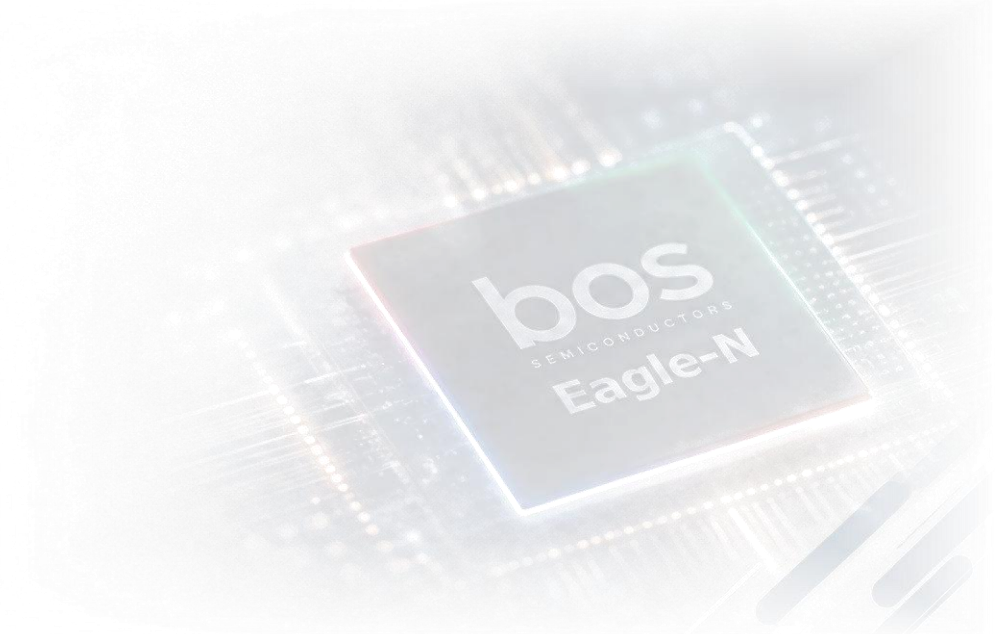
03 Core Remapping – The Cost of Data Movement

Disconnected Placements Require More Multicast Work



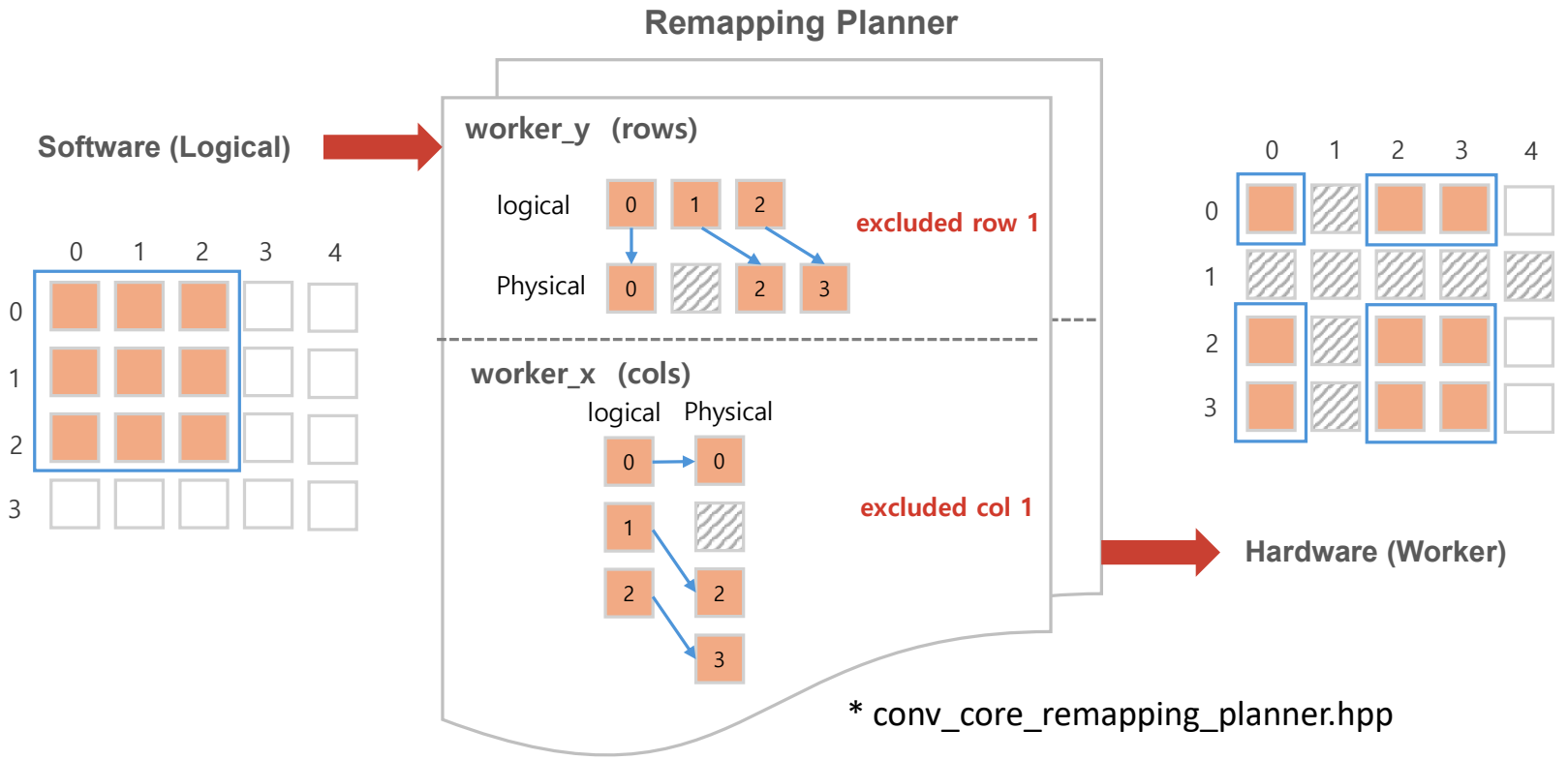
Each additional region repeats data and semaphore multicast, while the write barrier remains shared.

04 Remapping Planner



04 Remapping planner – A Placement Translation Layer

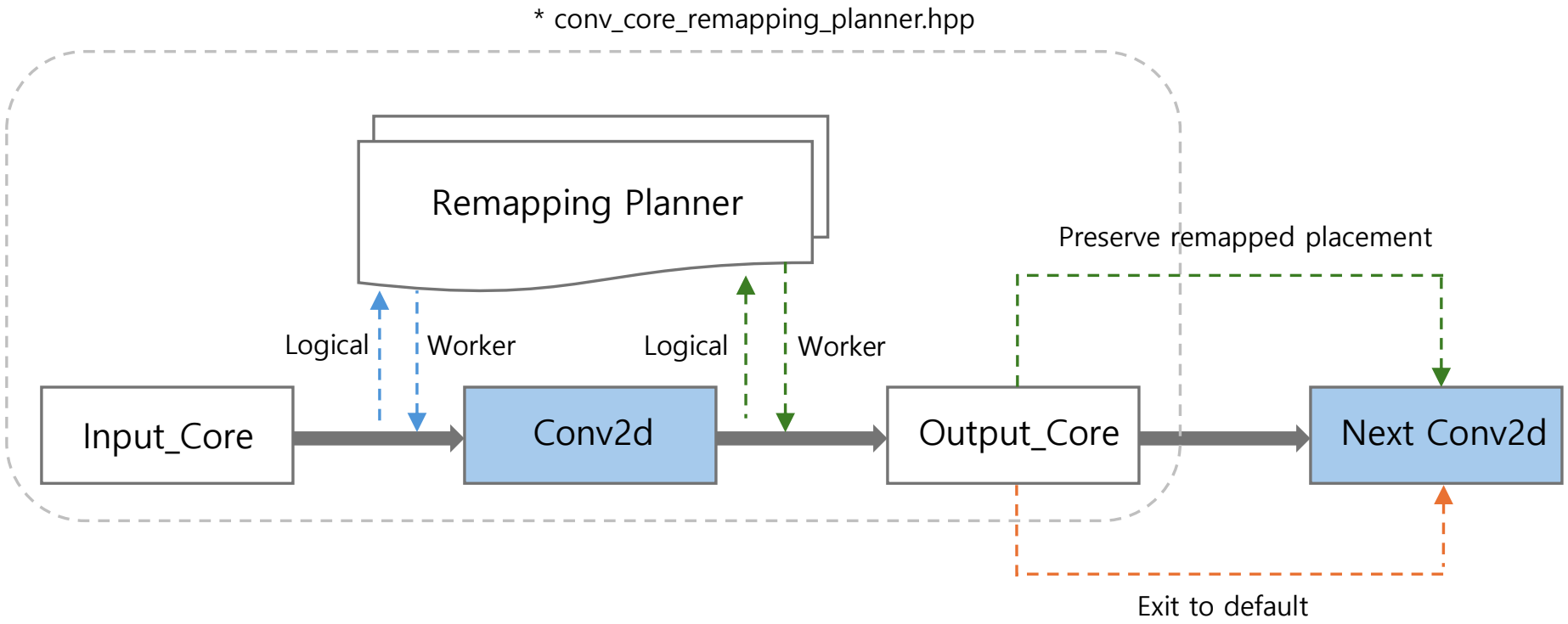
Axis-Wise Mapping Preserves Logical Shard Order



The planner isolates physical worker placement from logical tensor sharding.

04 Remapping planner – A Placement Translation Layer

The Planner Decides Whether to Keep or Reset the Placement



The remapped placement continues when possible; otherwise, it returns to the default placement.

04 Remapping Planner – **Programmable** Mechanism, **Policy**-Driven Choice

Expose placement flexibility without hard-coding a single cost trade-off.

Programmable API

1. Expose row and column exclusions

2. Support contiguous and segmented placements

3. Preserve shard order across operations

Placement Policy

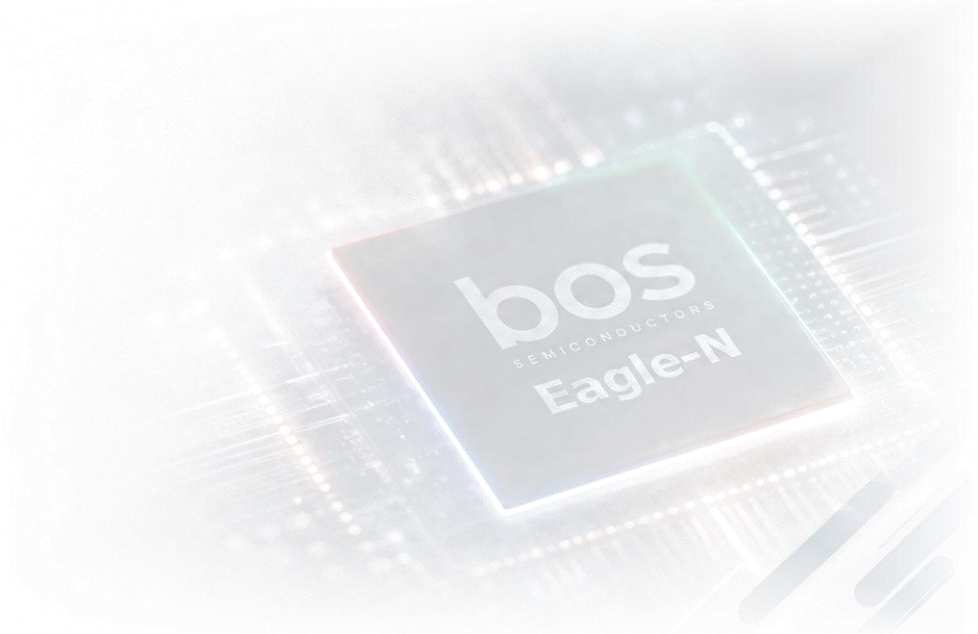
1. Balance compute exposure

2. Account for multicast fragmentation

3. Control placement changes across operations

The API defines what is possible. The policy decides what is worthwhile.

05 MILP Optimization



05 MILP Optimization – Mixed-Integer Linear Programming

A mathematical framework for discrete decisions under linear constraints.

MILP

Mixed — continuous and integer variables

Integer — discrete choices

Linear — linear objectives and constraints

Programming — mathematical optimization

$$\min_x \quad c^T x$$ **Objective**

$$\text{s. t.} \quad Ax \leq b$$

$$Ex = f$$ **Constraints**

$$\ell \leq x \leq u$$

$$x_i \in \mathbb{Z}, \quad i \in \mathcal{I}$$ **Discrete Decisions**

Constraints define what is feasible. The objective determines what is best.

05 MILP Optimization – Objectives in Priority Order

Each objective is optimized without giving up the result of the previous one.

$$L_c = HB_c + \sum_{d \in D} \sum_{p \in P_d^{(K)}} n_{dp} A_{dpc}$$

Total TRISC cycles accumulated on worker **c**

- B_c : Load from operations whose placement remains unchanged
- A_{dpc} : Load assigned to worker **c** when placement pis selected for operation **d**
- n_{dp} : Number of times placement **p** is selected for operation **d**
- $H = 20$: Number of inference repetitions

05 MILP Optimization – Objectives in Priority Order

Each step preserves the optimum achieved by earlier steps.

STEP 1 — Minimize Peak Load

$$\min z \qquad \text{s.t.} \qquad L_c \leq z, \quad \forall c$$

Reduce the load on the most heavily used worker.

STEP 2 — Balance Worker Loads

$$\min \sum_c | L_c - \bar{L} |$$

Reduce load variation across all workers.

STEP 3 — Reduce Placement Complexity

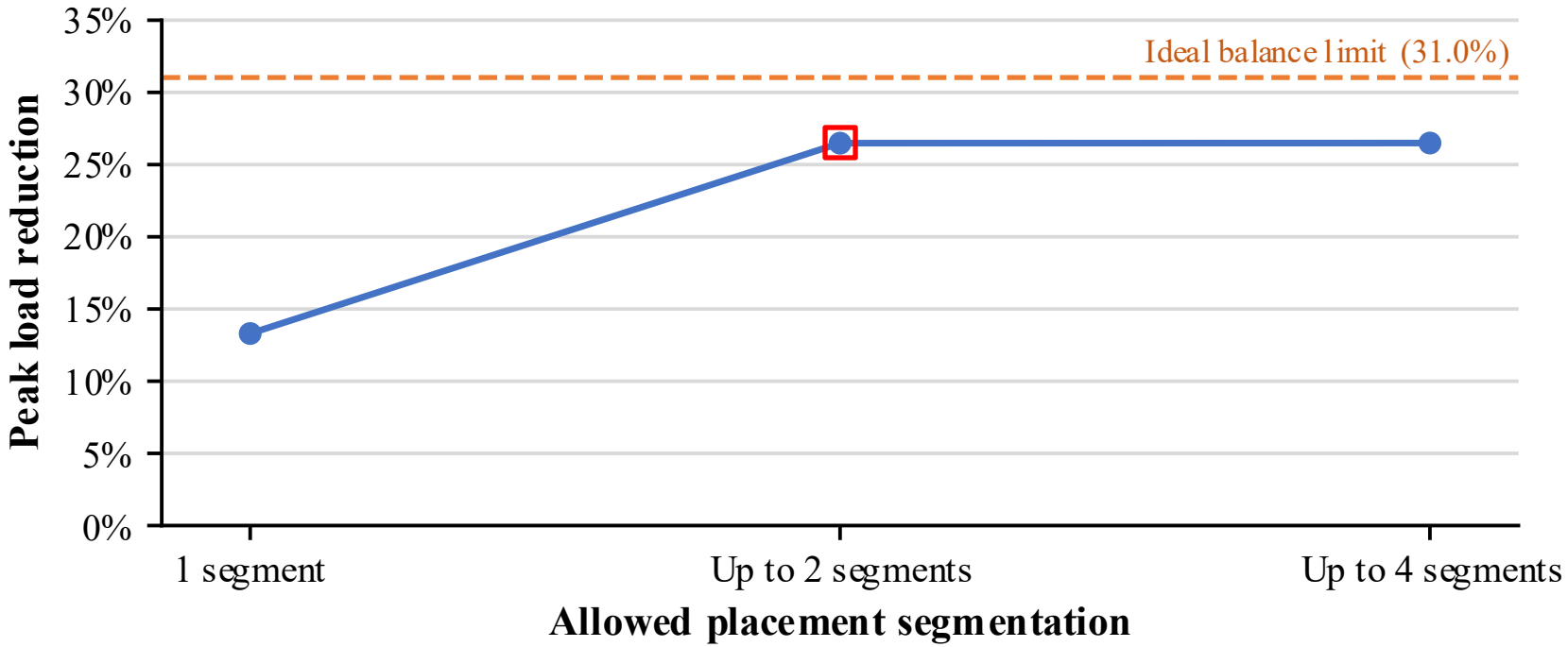
$$\min \sum_{d,p} u_{dp}$$

$$u_{dp} = 1 \quad \text{if placement } p \text{ is used}$$

Use fewer placement choices across operations.

05 MILP Optimization – Placement Flexibility vs. Peak Load

Two Segments Capture Most of the Achievable Improvement

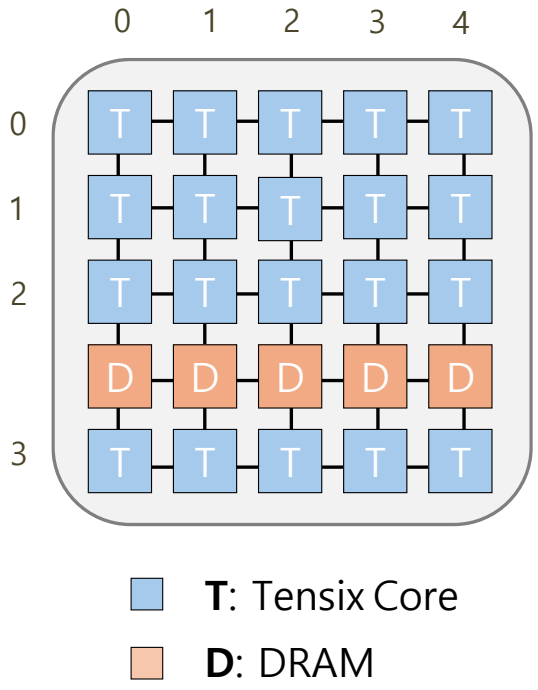


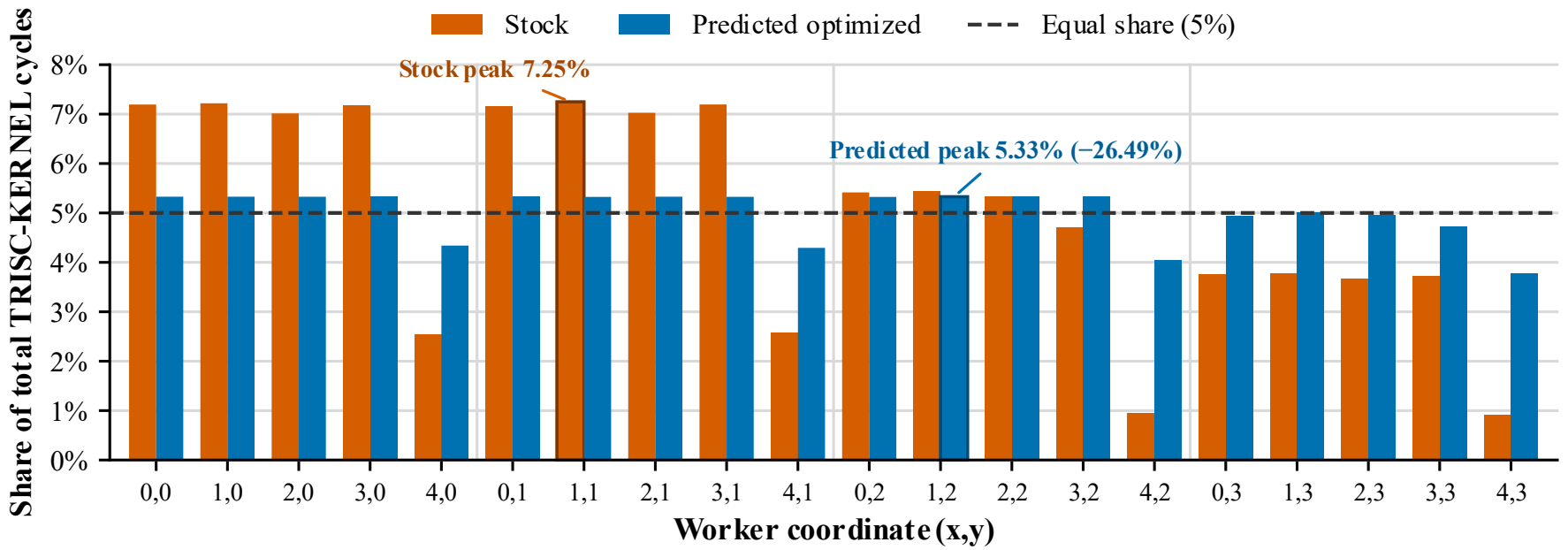
Solver-certified gap: 0.000989%

Allowing up to two segments increases peak-load reduction from 13.3% to 26.5%. Four segments provide no further gain, reaching 85.5% of the ideal balance limit.

05 MILP Optimization – Predicted Load Redistribution

Peak Worker Load Is Reduced Across the 20-Worker Grid

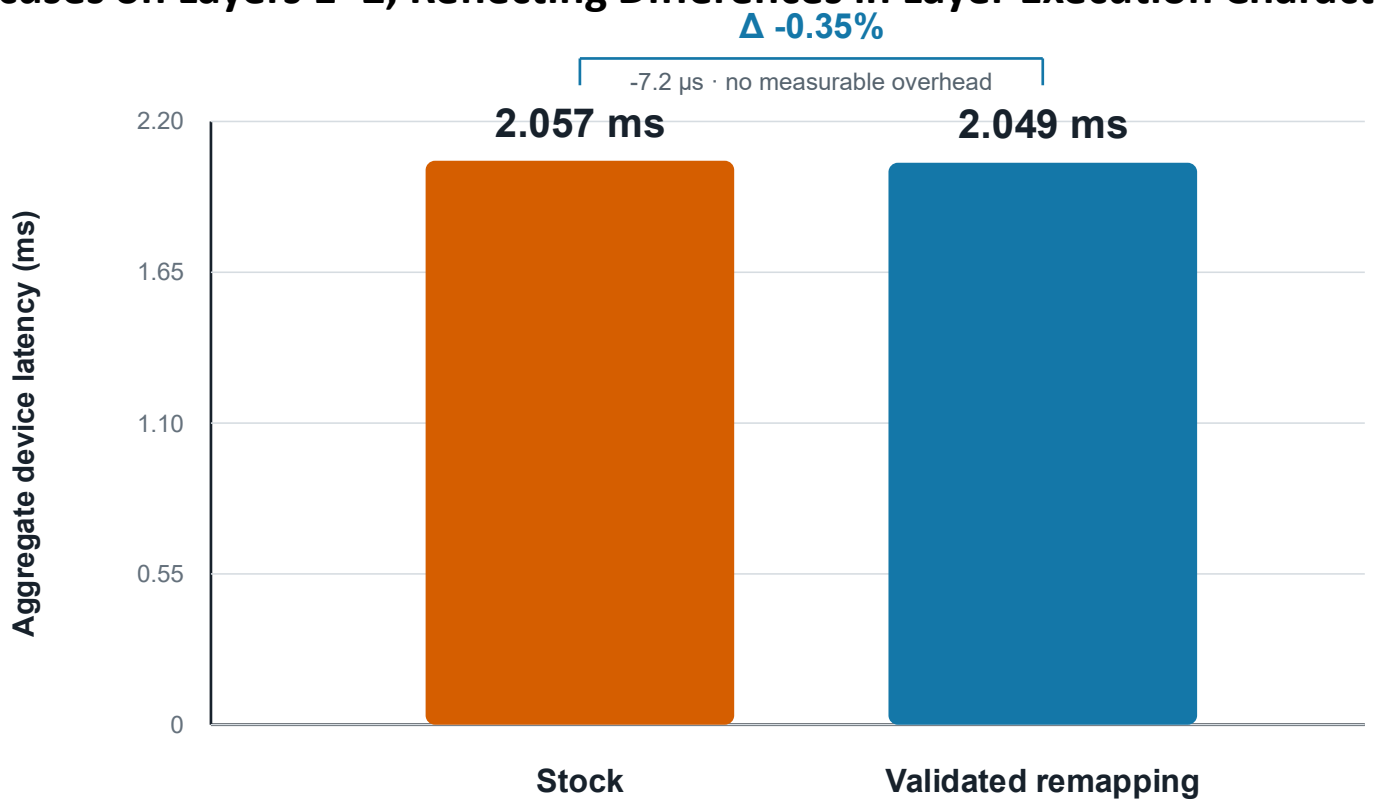




The optimized placements redistribute the same total TRISC work across all 20 workers.

06 Evaluation & Conclusion

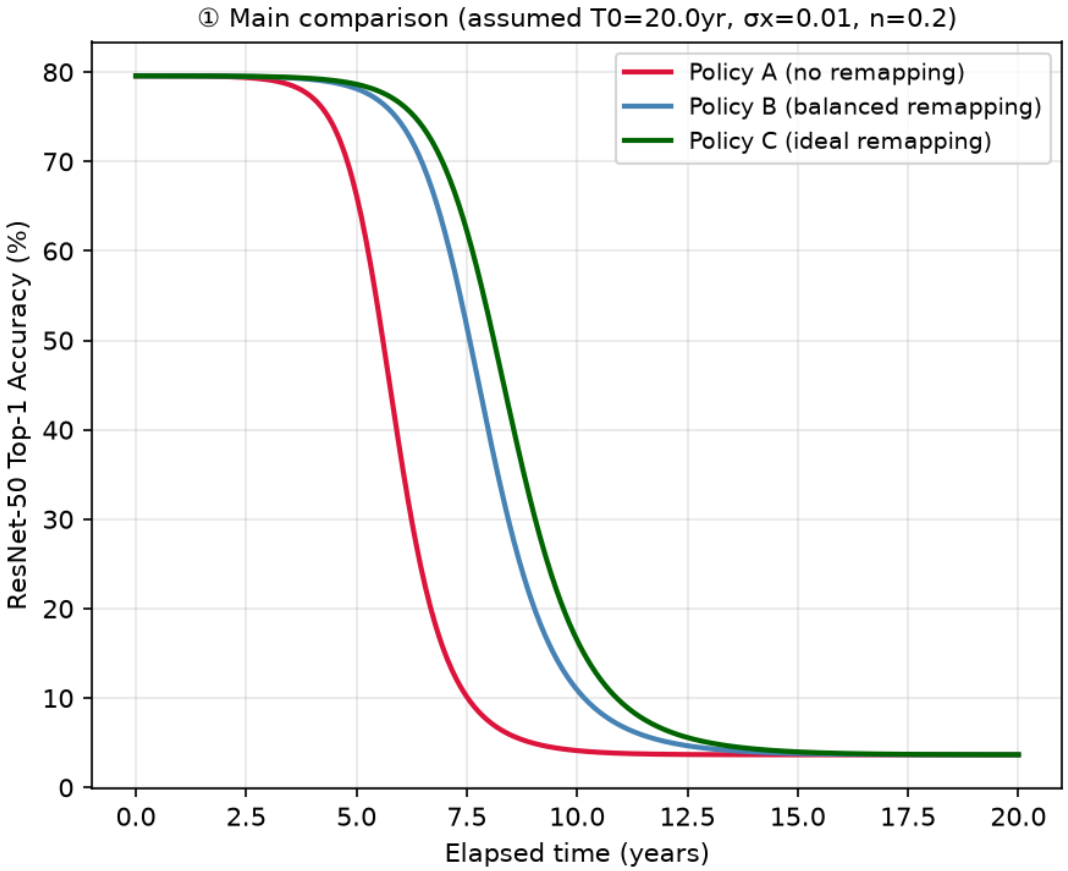
Initial Optimization Focuses on Layers 1–2, Reflecting Differences in Layer Execution Characteristics



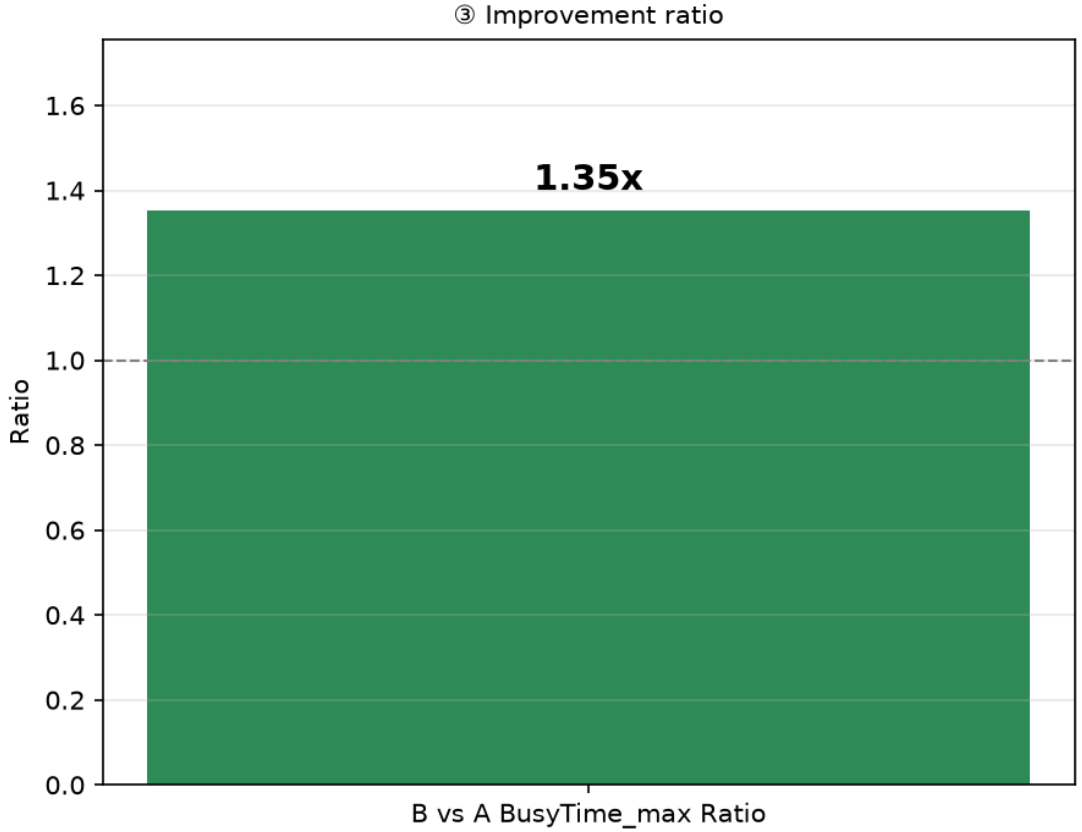
The right placement policy makes core remapping effectively overhead-free.

06 Evaluation & Conclusion

- Result



~ 1.5 years of lifetime improvement



1.35x improvement in stress time

Limitaions & Future work

Aging-Aware NPU Remapping

bos SEMICONDUCTORS x 성균관대학교 SUNG KYUN KWAN UNIVERSITY

Limitations: relative aging comparison only.

01 **Proxy ≠ Lifetime**

• Relative aging comparison only; not absolute values.

02 **Missing Telemetry**

• Approximations and assumptions due to process parameters and measurement limitations.

03 **Model Scope**

• Assumes purely computational operations and considers only NBTI effects.

04 **Policy Validation**

• Further verification required for performance and system overhead.

References

Standards, platform materials, and reliability-modeling sources used in this study.

Standards & Platform

[1] AEC, AEC-Q100 Rev. J1, Failure Mechanism Based Stress Test Qualification for Integrated Circuits in Automotive Applications, 2026.

[2] NHTSA, Part 573 Safety Recall Report 20V-369, Toyota Motor Engineering & Manufacturing, 2020.

[3] BOS Semiconductors, “Eagle-N, AI Accelerator for ADAS & IVI.”

[4] Tenstorrent, TT-Metalium Documentation, “Compute Engines and Data Flow within Tensix.”

Reliability & Statistical Modeling

[5] X. Yang et al., “Development and Challenges of Reliability Modeling From Transistors to Circuits,” IEEE Journal of the Electron Devices Society, vol. 11, pp. 179–189, 2023.

[6] H. Huang et al., “Statistical Modeling of Soft Error Influence on Neural Networks,” IEEE TCAD, vol. 42, no. 11, pp. 4152–4163, Nov. 2023, doi: 10.1109/TCAD.2023.3266405.

[7] A. Agarwal, D. Blaauw and V. Zolotov, “Statistical timing analysis for intra-die process variations with spatial correlations,” ICCAD, 2003, pp. 900–907, doi: 10.1109/ICCAD.2003.159781.

[8] V. Veetil, Y.-H. Chang, D. Sylvester and D. Blaauw, “Efficient smart monte carlo based SSTA on graphics processing units with improved resource utilization,” DAC, 2010, pp. 793–798, doi: 10.1145/1837274.1837474.

Thank you

